ASSIGNMENT : 10

TITLE : Write a Java program to demonstrate the use of abstract classes and abstract methods.

Problem Statement : Write a Java program to demonstrate the use of abstract classes and abstract methods.

Learning Objectives : To implement abstraction using abstract classes and abstract methods.

Theory : Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.

Exercise :

1. Create an abstract **Payment class** and implement Credit Card and UPI payment classes.

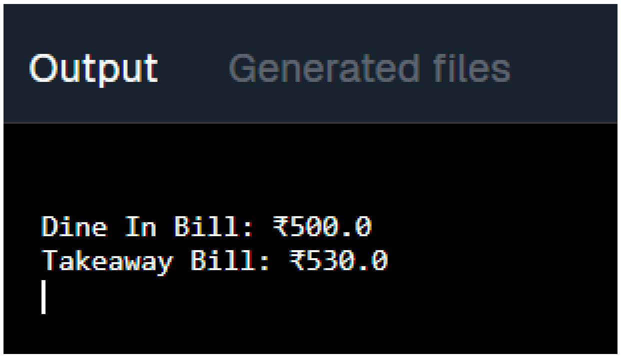
```
abstract class Payment {

    abstract void pay(double amount);
}

class CreditCard extends Payment {

    void pay(double amount) {
        System.out.println("Paid ₹" + amount + " using Credit Card.");
    }
}

class UPI extends Payment {

    void pay(double amount) {
        System.out.println("Paid ₹" + amount + " using UPI.");
    }
}

public class Main {
    public static void main(String[] args) {

        Payment p1 = new CreditCard();
        Payment p2 = new UPI();

        p1.pay(2500);
        p2.pay(1200);
    }
}
```

Output Generated files

```
Paid ₹2500.0 using Credit Card.
Paid ₹1200.0 using UPI.
```

2. Create an abstract **FoodOrder** class with an abstract method calculateBill(). Implement two subclasses, **DineInOrder** and **TakeAwayOrder**, to calculate and display the total bill based on the order type.

```java
abstract class FoodOrder {

    double amount;

    FoodOrder(double amount) {
        this.amount = amount;
    }

    abstract void calculateBill();
}

class DineInOrder extends FoodOrder {

    DineInOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        System.out.println("Dine In Bill: ₹" + amount);
    }
}

class TakeAwayOrder extends FoodOrder {

    TakeAwayOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        System.out.println("Takeaway Bill: ₹" + (amount + 30));
    }
}

public class Main {
    public static void main(String[] args) {

        FoodOrder d = new DineInOrder(500);
        FoodOrder t = new TakeAwayOrder(500);

        d.calculateBill();
        t.calculateBill();
    }
}
```



Outcome :

- Understood the concepts of abstract classes and abstract methods.
- Implemented abstraction by defining common behavior in abstract classes.
- Learned how subclasses provide specific implementations of abstract methods.

Conclusion :

This assignment demonstrated abstraction using abstract classes and abstract methods. Abstract classes provide a common framework while allowing subclasses to implement specific functionality, resulting in flexible, reusable, and maintainable Java programs.